

TRELLIS.2 Printability Fine-Tuning - Weekly Report

1. Executive summary

Work this period followed three stages: improving printability after generation, attempting to steer generation itself with DreamDPO concepts, and finally fine-tuning the SLat flow model directly.

Post-processing was addressed first. Many post-processing libraries and methods were tried to improve quality and printability, and different options were added to the pipeline. Of those, manifold3d proved the best for 3D printing, and it was added while keeping the original trimesh pipeline in place. Further research was done on the best way to decrease non-manifold edges of the models, and the TriFlow paper on generating artist-like 3D models was read as part of that.

Integrating DreamDPO concepts was attempted and did not succeed. The DreamDPO work for diffusion models was read, and rather than using a VLM, the preference discriminator was simulated as a score. That produced no gains: it always chose the reference path for the flow model. Combined with a lack of expertise in this area of the theory, the approach was set aside.

Fine-tuning the SLat flow model with LoRA was chosen instead, tried on a small sample of 300 curated 3D models from Thingi10K. Initial training showed gains, with significantly fewer non-manifold edges on the raw results and without loss in detail. Collection of more latent data for training is continuing.

Measured on exported meshes, held-out set, paired sampling noise:

Model	L_OH (overhang)	L_Th (thin walls)	L_Topo	R_Detail
Base model	0.2291	0.1133	0.1871	0.4316
step 750	0.2183 (-5%)	0.0152 (-87%)	0.0500 (-73%)	0.4578 (+6%)
step 900	0.2255 (-2%)	0.0299 (-74%)	0.0635 (-66%)	0.4411 (+2%)
step 1050	0.2179 (-5%)	0.0159 (-86%)	0.0340 (-82%)	0.4358 (+1%)

Step 1050 beats the base model on thin-wall penalty for 6 of 6 held-out objects. A separate paired test over 10 objects and 3 sampling seeds found non-manifold edges reduced by 82% against the base model, winning on 10 of 10 objects. Detail was not lost: R_Detail rose slightly and image fidelity did not degrade (0.897 -> 0.908-0.913).

2. What was discovered

2.1 The retired approach failed for a diagnosable reason

Inference-time DPO steering produced a gradient contribution of -0.0027 +/- 0.0078 over 17 scored forks - indistinguishable from zero. Root cause: the only differentiable scalar available (slat_detail_proxy) agreed with the judge's within-fork argmax 5.9% of the time against a 33% chance line, with mean Spearman -0.265. The optimiser worked; its objective was near-orthogonal to the goal.

2.2 The "differentiability wall" did not exist

2.3 The VAE bounds what any flow-model training can achieve

Property	Result
Shape fidelity	Preserved - Chamfer 0.0052, p95 0.0101, bounding box exact
Watertightness	0 of 2 preserved
Defect rate (full mesh)	0.086% open edges, 0.095% non-manifold
Components	953, but largest is 99.85% of faces - and still not watertight alone

Consequence: geometry-carried printability (thickness, overhang, bulk shape) is reachable by training the flow model; watertightness is not, and belongs to post-processing. This single 20-minute measurement redirected the entire project.

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Encoding a

2.4 The judge inflates topology defects ~50x

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2.5 Measurement noise nearly made the whole run uninterpretable

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2.6 Silent failures dominated the engineering cost

Every sign

Symptom	Actual cause	Cost
Training step took 464s	No activation checkpointing -> 38 GB retained	18x slowdown
Baseline eval returned n_eval=0	Three stacked bugs, masked by except: contin	2 invalidated launches
Eval decoder on CPU	low_vram is device placement, not a memory hi	1 launch
Sample-evolution panel empty	A patch silently failed to match; the call site was	whole run, cosmetic
Loss chart nonsense	Dataset build logged token counts into the loss f	misleading chart

The single highest-leverage change all week was replacing except Exception: continue with recorded, printed failures. It converted two blind retries into one-step diagnoses.

2.7 Post-processing was addressed before generation was

Many post-
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was added
best way to
was read.

This matters for reading the rest of this report: the fine-tuning results quoted here are on raw decoder output, with no post-processing applied. The reduction in non-manifold edges is therefore a gain made before manifold3d or trimesh repair runs, not instead of them. Both remain in the pipeline, and the round-trip result in sec.2.3 is why - watertightness is a decoder property that fine-tuning cannot reach.

3. What was built

Component	File	Note
CPU mesh -> SLat encoder	backends/dual_grid_cpu.py	o-voxel ships CUDA-only; rebuilt the pure-C++
Offscreen renderer	trellis_core/render_mesh.py	moderngl, Metal-backed, headless; camera fitt
Dataset pipeline	trellis_core/dataset_build.py	300 models, 3.2 GPU-hours, 0 failures
LoRA	trellis_core/lora.py	210 modules, 18.68M params (1.4% of 1.29B)
SFT trainer	trellis_core/sft_train.py	Hyperparameters from TRELLIS.2's own config
Live dashboard	devlab/train_server.py	Crash-independent; phone access over Tailsca
VAE round-trip test	trellis_core/vae_roundtrip.py	Established the ceiling
Mesh exporter	trellis_core/export_samples.py	Reproduces eval exactly (same seeds, same
Topology test	trellis_core/topology_test.py	Paired, undecimated, multi-seed

Dataset

Thing10K
3D-printing

multi-comp
PWN + no
(270 train /

Training run

steps 1,170 of 1,200 (stopped after divergence)
epochs 8.7 over 270 examples
GPU time 9.96 hours, mean 30.7 s/step
LoRA rank 16, alpha 32, lr 1e-4 constant, AdamW wd 0.01
schedule uniform t, p_uncond 0.1 (both from TRELLIS.2's official config)
evals 10, every 150 steps, 6 held-out objects, common random numbers

run id

4. The divergence

Steps	Mean loss	Max loss	Grad norm
1-1100	~0.16	0.80	0.03-0.06
1101-1170	3.48	57.29	332.25

Held-out loss went 0.0859 -> 10.29; image similarity 0.908 -> 0.202. The model was destroyed.

Two causes, both mine: no learning-rate decay (constant 1e-4 for 8.7 epochs), and gradient clipping that bounded each update but could not prevent AdamW's moments accumulating through a sustained blow-up. Checkpoints 750/900/1050 were preserved before stopping, because a clean stop writes a final checkpoint and prune_checkpoints(keep=3) would have deleted step 750.

5. How to log improvements numerically

Recommended standard for every future comparison. The first three are non-negotiable given what we measured this week.

5.1 Always paired, never pooled

Fix the sam
per-object
is between
within-fork

5.2 Always against a baseline

Every run l
number wi

5.3 Always report fidelity beside printability

Minimum p
condition
experimen

5.4 The metric set

				Metric
L_Th	judge	lower	Responds strongly. Primary succe	
nonmanifold_rate, open_rate	full mesh	lower	Must be computed undecimated (	
L_OH	judge	lower	Has not responded; also only ~5%	
R_Detail	judge	context	Fidelity term, not printability - no be	
image_similarity	silhouette IoU	higher	Prompt-adherence guard	
sample_dispersion	latent means	flat	Collapse guard	
watertight_rate	full mesh	control	Expected flat; a change indicates a	
support volume	slicer, not yet built	lower	The only metric outside the trainin	

5.5 The acceptance bar

Improvement

```
capture = (tuned_1_sample - base_1_sample) / (base_best_of_N - base_1_sample)
```

A single sample from the tuned model must recover a meaningful fraction of what N-fold rejection sampling buys from the base model at matched compute. This test has not yet been run.

6. Next steps

1. Finish the topology consistency test (running: 10 objects x 3 seeds x 4 models, paired, undecimated). Answers whether step 1050's cleaner topology is consistent or a lucky draw.
2. Run the best-of-N comparison (sec.5.5). Until this exists we do not know whether the fine-tune beats cheap rejection sampling - the only comparison that justifies the work.
3. Fix the training schedule and re-run. Cosine LR decay; a skip-step rule that drops a batch whose gradient norm exceeds a multiple of the running median rather than clipping it; keep every checkpoint. The metrics were still improving when the run broke, so the ceiling was never observed.
4. Repair the judge before it is trusted further: make the thickness raycast deterministic (test-retest reliability was measured at 0.208, capping any correlation at $|r| \leq 0.456$), and add a support raycast to the overhang term (currently rank-correlates -0.432 with true unsupported area).
5. Scale the dataset. 270 training objects is the binding constraint, not step count; 3,935 clean models are available and prep runs at ~40 s/model.
6. Add a slicer-based support-volume metric as the held-out physical measure that was never part of the training reward.

7. Deliverables on disk

runs/sft-1200-20260806-0119/	
ckpt_keep/step_{750,900,1050}/adapter.pt	preserved LoRA adapters (643 MB)
meshes/step_{0,750,900,1050}/	24 GLB + 24 STL, raw, no post-processing
exports/	42 PNG frames + 6 progression strips
metrics.jsonl	1,170 train + 10 eval records
topology_test.jsonl	in progress
data/thingi10k_sft/	300 latents + images + conditioning

All meshes are raw decoder output - no repair, no decimation, no solidify. Each object uses identical conditioning and identical sampling noise across all four models, so differences are attributable to the model alone.